

Week 10 — Day 1

Django Migration Lab — Complete Guide + Screenshots

This English guide records the exact lab flow, what was added or changed at each stage, the commands used, and the terminal results. Screenshots are placed with the relevant step.

Project: migration_lab App: catalog Database: SQLite Django: 6.1.1

1. Project Setup

Create the project folder, virtual environment, install Django, create the Django project and catalog app.

```
mkdir migration_lab cd migration_lab python -m venv venv venv\Scripts\activate pip install django django-admin --version django-admin startproject migration_project . python manage.py check python manage.py startapp catalog code .
```

```
C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1>mkdir migration_lab
C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1>cd migration_lab
C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python -m venv venv
C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>venv\Scripts\activate
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>pip install django
Collecting django
  Using cached django-6.1.1-py3-none-any.whl.metadata (3.9 kB)
Collecting asgiref<=3.9.1 (from django)
  Using cached asgiref-3.12.1-py3-none-any.whl.metadata (9.4 kB)
Collecting sqlparse<=0.5.0 (from django)
  Using cached sqlparse-0.6.0-py3-none-any.whl.metadata (6.0 kB)
Collecting tzdata (from django)
  Using cached tzdata-2026.4-py2.py3-none-any.whl.metadata (1.4 kB)
Using cached django-6.1.1-py3-none-any.whl (8.4 MB)
Using cached asgiref-3.12.1-py3-none-any.whl (25 kB)
Using cached sqlparse-0.6.0-py3-none-any.whl (50 kB)
Using cached tzdata-2026.4-py2.py3-none-any.whl (347 kB)
Installing collected packages: tzdata, sqlparse, asgiref, django
Successfully installed asgiref-3.12.1 django-6.1.1 sqlparse-0.6.0 tzdata-2026.4
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>django-admin --version
6.1.1
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>django-admin startproject migration_project .
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py check
System check identified no issues (0 silenced).
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py startapp catalog
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>code .
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py check
System check identified no issues (0 silenced).
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog
No changes detected in app 'catalog'
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog
No changes detected in app 'catalog'
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog
Migrations for 'catalog':
  catalog\migrations\0001_initial.py
  + Create model Product
```

Project creation, Django installation, checks, and catalog app creation.

2. Register catalog in settings.py

Add 'catalog' to INSTALLED_APPS so Django discovers the app's models and migrations.

```
'catalog',
```

Application definition

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'catalog',  
]
```

catalog is registered in INSTALLED_APPS.

3. Create the Initial Product Model

The first Product model contains name and stock.

```
class Product(models.Model): name = models.CharField(max_length=100) stock = models.IntegerField(default=0)  
python manage.py makemigrations catalog
```

This creates 0001_initial.py.

```
C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1>mkdir migration_lab  
C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1>cd migration_lab  
C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python -m venv venv  
C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>venv\Scripts\activate  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>pip install django  
Collecting django  
  Using cached django-6.1.1-py3-none-any.whl.metadata (3.9 kB)  
Collecting asgiref<=3.9.1 (from django)  
  Using cached asgiref-3.12.1-py3-none-any.whl.metadata (9.4 kB)  
Collecting sqlparse<=0.5.0 (from django)  
  Using cached sqlparse-0.6.0-py3-none-any.whl.metadata (6.0 kB)  
Collecting tzdata (from django)  
  Using cached tzdata-2026.4-py2.py3-none-any.whl.metadata (1.4 kB)  
Using cached django-6.1.1-py3-none-any.whl (8.4 MB)  
Using cached asgiref-3.12.1-py3-none-any.whl (25 kB)  
Using cached sqlparse-0.6.0-py3-none-any.whl (50 kB)  
Using cached tzdata-2026.4-py2.py3-none-any.whl (347 kB)  
Installing collected packages: tzdata, sqlparse, asgiref, django  
Successfully installed asgiref-3.12.1 django-6.1.1 sqlparse-0.6.0 tzdata-2026.4  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>django-admin --version  
6.1.1  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>django-admin startproject migration_project .  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py check  
System check identified no issues (0 silenced).  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py startapp catalog  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>code .  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py check  
System check identified no issues (0 silenced).  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog  
No changes detected in app 'catalog'  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog  
No changes detected in app 'catalog'  
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog  
Migrations for 'catalog':  
  catalog\migrations\0001_initial.py  
    + Create model Product
```

0001_initial.py is created for the Product model.

4. Add Product.code as Nullable

Add the new code field as nullable first. This lets existing database rows remain valid while the new column is introduced.

```
code = models.CharField(max_length=30, null=True)

python manage.py makemigrations catalog --name add_product_code
```

This creates 0002_add_product_code.py.

```
(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog
Migrations for 'catalog':
  catalog\migrations\0002_product_code.py
  + Add field code to product

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations
admin
[ ] 0001_initial
[ ] 0002_logentry_remove_auto_add
[ ] 0003_logentry_add_action_flag_choices
auth
[ ] 0001_initial
[ ] 0002_alter_permission_name_max_length
[ ] 0003_alter_user_email_max_length
[ ] 0004_alter_user_username_opts
[ ] 0005_alter_user_last_login_null
[ ] 0006_require_contenttypes_0002
[ ] 0007_alter_validators_add_error_messages
[ ] 0008_alter_user_username_max_length
[ ] 0009_alter_user_last_name_max_length
[ ] 0010_alter_group_name_max_length
[ ] 0011_update_proxy_permissions
[ ] 0012_alter_user_first_name_max_length
catalog
[ ] 0001_initial
[ ] 0002_product_code
contenttypes
[ ] 0001_initial
[ ] 0002_remove_content_type_name
sessions
[ ] 0001_initial

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate --plan
Planned operations:
contenttypes.0001_initial
  Create model ContentType
  Alter unique_together for contenttype (1 constraint(s))
auth.0001_initial
  Create model Permission
  Create model Group
  Create model User
admin.0001_initial
  Create model LogEntry
admin.0002_logentry_remove_auto_add
  Alter field action_time on logentry
admin.0003_logentry_add_action_flag_choices
  Alter field action_flag on logentry
contenttypes.0002_remove_content_type_name
  Change Meta options on contenttype
  Alter field name on contenttype
  Raw Python operation
  Remove field name from contenttype
auth.0002_alter_permission_name_max_length
  Alter field name on permission
auth.0003_alter_user_email_max_length
  Alter field email on user
auth.0004_alter_user_username_opts
  Alter field username on user
auth.0005_alter_user_last_login_null
  Alter field last_login on user
auth.0006_require_contenttypes_0002
  Alter field username on user
auth.0007_alter_validators_add_error_messages
  Alter field username on user
auth.0008_alter_user_username_max_length
  Alter field username on user
auth.0009_alter_user_last_name_max_length
  Alter field last_name on user
```

0002_add_product_code.py is created and catalog migrations are listed.

5. Inspect the Migration

```
python manage.py showmigrations catalog python manage.py migrate --plan python manage.py sqlmigrate catalog 0002
```

showmigrations checks migration status, migrate --plan previews operations, and sqlmigrate shows the SQL Django generated.

```

auth.0007_alter_validators_add_error_messages
    Alter field username on user
auth.0008_alter_user_username_max_length
    Alter field username on user
auth.0009_alter_user_last_name_max_length
    Alter field last_name on user
auth.0010_alter_group_name_max_length
    Alter field name on group
auth.0011_update_proxy_permissions
    Raw Python operation -> Update the content_type of proxy mo...
auth.0012_alter_user_first_name_max_length
    Alter field first_name on user
catalog.0001_initial
    Create model Product
catalog.0002_product_code
    Add field code to product
sessions.0001_initial
    Create model Session

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>del catalog\migrations\0002_product_code.py

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog --name add_product_code
Migrations for 'catalog':
  catalog\migrations\0002_add_product_code.py
    + Add field code to product

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
catalog
[ ] 0001_initial
[ ] 0002_add_product_code

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate --plan
Planned operations:
contenttypes.0001_initial
    Create model ContentType
    Alter unique_together for contenttype (1 constraint(s))
auth.0001_initial
    Create model Permission
    Create model Group
    Create model User
admin.0001_initial
    Create model LogEntry
admin.0002_logentry_remove_auto_add
    Alter field action_time on logentry
admin.0003_logentry_add_action_flag_choices
    Alter field action_flag on logentry
contenttypes.0002_remove_content_type_name
    Change Meta options on contenttype
    Alter field name on contenttype
    Raw Python operation
    Remove field name from contenttype
auth.0002_alter_permission_name_max_length
    Alter field name on permission
auth.0003_alter_user_email_max_length
    Alter field email on user
auth.0004_alter_user_username_opts
    Alter field username on user
auth.0005_alter_user_last_login_null
    Alter field last_login on user
auth.0006_require_contenttypes_0002
    Alter field username on user
auth.0007_alter_validators_add_error_messages
    Alter field username on user
auth.0008_alter_user_username_max_length
    Alter field username on user
auth.0009_alter_user_last_name_max_length
    Alter field last_name on user
auth.0010_alter_group_name_max_length
    Alter field name on group

```

sqlmigrate shows the SQL for adding the nullable code column.

6. Create the Data Migration

Create an empty migration named `fill_product_codes` to populate codes for existing Product rows.

```
python manage.py makemigrations catalog --empty --name fill_product_codes
```

The migration uses `RunPython`. A forward function fills codes and a reverse function can clear them.

```
def fill_codes(apps, schema_editor): Product = apps.get_model("catalog", "Product") for product in
Product.objects.iterator(): product.code = f"P{product.pk:06d}" product.save(update_fields=["code"]) def
clear_codes(apps, schema_editor): Product = apps.get_model("catalog", "Product")
Product.objects.update(code=None)
```

```
python manage.py migrate
```

In the lab, the built-in Django migrations plus catalog 0001, 0002, and 0003 completed successfully.

```

auth.0007_alter_validators_add_error_messages
    Alter field username on user
auth.0008_alter_user_username_max_length
    Alter field username on user
auth.0009_alter_user_last_name_max_length
    Alter field last_name on user
auth.0010_alter_group_name_max_length
    Alter field name on group
auth.0011_update_proxy_permissions
    Raw Python operation -> Update the content_type of proxy mo...
auth.0012_alter_user_first_name_max_length
    Alter field first_name on user
catalog.0001_initial
    Create model Product
catalog.0002_product_code
    Add field code to product
sessions.0001_initial
    Create model Session

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>del catalog\migrations\0002_product_code.py

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog --name add_product_code
Migrations for 'catalog':
  catalog\migrations\0002_add_product_code.py
    + Add field code to product

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
catalog
[ ] 0001_initial
[ ] 0002_add_product_code

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate --plan
Planned operations:
contenttypes.0001_initial
    Create model ContentType
    Alter unique_together for contenttype (1 constraint(s))
auth.0001_initial
    Create model Permission
    Create model Group
    Create model User
admin.0001_initial
    Create model LogEntry
admin.0002_logentry_remove_auto_add
    Alter field action_time on logentry
admin.0003_logentry_add_action_flag_choices
    Alter field action_flag on logentry
contenttypes.0002_remove_content_type_name
    Change Meta options on contenttype
    Alter field name on contenttype
    Raw Python operation
    Remove field name from contenttype
auth.0002_alter_permission_name_max_length
    Alter field name on permission
auth.0003_alter_user_email_max_length
    Alter field email on user
auth.0004_alter_user_username_opts
    Alter field username on user
auth.0005_alter_user_last_login_null
    Alter field last_login on user
auth.0006_require_contenttypes_0002
    Alter field username on user
auth.0007_alter_validators_add_error_messages
    Alter field username on user
auth.0008_alter_user_username_max_length
    Alter field username on user
auth.0009_alter_user_last_name_max_length
    Alter field last_name on user
auth.0010_alter_group_name_max_length
    Alter field name on group

```

0003_fill_product_codes is created and the migration run completes successfully.

7. Make code Required and Unique

After existing rows have valid codes, change the model field to:

```
code = models.CharField(max_length=30, unique=True)
```

When makemigrations warns that a nullable field cannot become non-nullable without handling existing rows, option 2 was selected because the data migration already handled the existing values.

```
python manage.py makemigrations catalog
```

Django creates 0004_alter_product_code.py.

```
python manage.py migrate --plan python manage.py sqlmigrate catalog 0004
```

```
Alter field name on group
auth.0011_update_proxy_permissions
Raw Python operation -> Update the content_type of proxy mo...
auth.0012_alter_user_first_name_max_length
Alter field first_name on user
catalog.0001_initial
Create model Product
catalog.0002_add_product_code
Add field code to product
sessions.0001_initial
Create model Session

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py sqlmigrate catalog 0002
BEGIN;
--
-- Add field code to product
ALTER TABLE "catalog_product" ADD COLUMN "code" varchar(30) NULL;
COMMIT;

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog --empty --name fill_product_codes
Migrations for 'catalog':
catalog/migrations/0003_fill_product_codes.py

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>code .

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate
Operations to perform:
Apply all migrations: admin, auth, catalog, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying catalog.0001_initial... OK
  Applying catalog.0002_add_product_code... OK
  Applying catalog.0003_fill_product_codes... OK
  Applying sessions.0001_initial... OK

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog
It is impossible to change a nullable field 'code' on product to non-nullable without providing a default. This is because the database needs something to populate existing rows.
Please select a fix:
  1) Provide a one-off default now (will be set on all existing rows with a null value for this column)
  2) Ignore for now. Existing rows that contain NULL values will have to be handled manually, for example with a RunPython or RunSQL operation.
  3) Quit and manually define a default value in models.py.
Select an option: 2
Migrations for 'catalog':
catalog/migrations/0004_alter_product_code.py
- Alter field code on product

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate --plan
Planned operations:
catalog.0004_alter_product_code
```

0004 is created and sqlmigrate shows the required + unique database change.

8. Rename stock to quantity

Change the model field name from stock to quantity.

```
quantity = models.IntegerField(default=0)
```

```
python manage.py makemigrations catalog python manage.py sqlmigrate catalog 0005
```

When Django asks whether stock was renamed to quantity, answer y. This creates 0005_rename_stock_product_quantity.py.

```

    Alter field name on group
auth.0011_update_proxy_permissions
Raw Python operation => Update the content_type of proxy mo...
auth.0012_alter_user_first_name_max_length
    Alter field first_name on user
catalog.0001_initial
    Create model Product
catalog.0002_add_product_code
    Add field code to product
sessions.0001_initial
    Create model Session

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py sqlmigrate catalog 0002
BEGIN;
--
-- Add field code to product
--
ALTER TABLE "catalog_product" ADD COLUMN "code" varchar(30) NULL;
COMMIT;

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog --empty --name fill_product_codes
Migrations for 'catalog':
  catalog\migrations\0003_fill_product_codes.py

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>code .

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, catalog, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying catalog.0001_initial... OK
  Applying catalog.0002_add_product_code... OK
  Applying catalog.0003_fill_product_codes... OK
  Applying sessions.0001_initial... OK

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog
It is impossible to change a nullable field 'code' on product to non-nullable without providing a default. This is because the database needs something to populate existing rows.
Please select a fix:
  1) Provide a one-off default now (will be set on all existing rows with a null value for this column)
  2) Ignore for now. Existing rows that contain NULL values will have to be handled manually, for example with a RunPython or RunSQL operation.
  3) Quit and manually define a default value in models.py.
Select an option: 2
Migrations for 'catalog':
  catalog\migrations\0004_alter_product_code.py
    ~ Alter field code on product

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate --plan
Planned operations:
  catalog.0004_alter_product_code

```

0005 is created and the generated SQL renames stock to quantity.

9. Apply All Migrations

python manage.py migrate python manage.py showmigrations catalog

The expected final catalog state is 0001 through 0005 all marked [X].

```

catalog.0004_alter_product_code
    Alter field code on product

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py sqlmigrate catalog 0004
BEGIN;
--
-- Alter field code on product
--
CREATE TABLE "new_catalog_product" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "code" varchar(30) NOT NULL UNIQUE, "name" varchar(100) NOT NULL, "stock" integer NOT NULL);
INSERT INTO "new_catalog_product" ("id", "name", "stock", "code") SELECT "id", "name", "stock", coalesce("code", NULL) FROM "catalog_product";
DROP TABLE "catalog_product";
ALTER TABLE "new_catalog_product" RENAME TO "catalog_product";
COMMIT;

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog
Was product.stock renamed to product.quantity (a IntegerField)? [y/N] y
Migrations for 'catalog':
  catalog\migrations\0005_rename_stock_product_quantity.py
    ~ Rename field stock on product to quantity

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate --plan
Planned operations:
  catalog.0004_alter_product_code
  catalog.0005_rename_stock_product_quantity
  ~ Rename field stock on product to quantity

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py sqlmigrate catalog 0005
BEGIN;
--
-- Rename field stock on product to quantity
--
ALTER TABLE "catalog_product" RENAME COLUMN "stock" TO "quantity";
COMMIT;

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, catalog, contenttypes, sessions
Running migrations:
  Applying catalog.0004_alter_product_code... OK
  Applying catalog.0005_rename_stock_product_quantity... OK

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
catalog
[X] 0001_initial
[X] 0002_add_product_code
[X] 0003_fill_product_codes
[X] 0004_alter_product_code
[X] 0005_rename_stock_product_quantity

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate catalog 0004
Operations to perform:
  Target specific migration: 0004_alter_product_code, from catalog
Running migrations:
  Rendering model states... DONE
  Unapplying catalog.0005_rename_stock_product_quantity... OK

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
catalog
[X] 0001_initial
[X] 0002_add_product_code
[X] 0003_fill_product_codes
[X] 0004_alter_product_code
[ ] 0005_rename_stock_product_quantity

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate catalog
Operations to perform:
  Apply all migrations: catalog
Running migrations:
  Applying catalog.0005_rename_stock_product_quantity... OK

(venv) C:\Users\sosyi\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>

```

All catalog migrations 0001-0005 are applied.

10. Roll Back to 0004

Target migration 0004 to test reversing 0005.

```
python manage.py migrate catalog 0004 python manage.py showmigrations catalog
```

Django un-applies 0005. The expected state is 0001-0004 [X] and 0005 [].

```
catalog.0004.alter_product_code
Alter field code on product

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py sqlmigrate catalog 0004
BEGIN;
--
-- Alter field code on product
--
CREATE TABLE "new_catalog_product" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "code" varchar(30) NOT NULL UNIQUE, "name" varchar(100) NOT NULL, "stock" integer NOT NULL);
INSERT INTO "new_catalog_product" ("id", "name", "stock", "code") SELECT "id", "name", "stock", coalesce("code", NULL) FROM "catalog_product";
DROP TABLE "catalog_product";
ALTER TABLE "new_catalog_product" RENAME TO "catalog_product";
COMMIT;

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
Was product.stock renamed to product.quantity (a IntegerField)? [y/N] y
Migrations for 'catalog':
  catalog\migrations\0005_rename_stock_product_quantity.py
  ~ Rename field stock on product to quantity

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate --plan
Planned operations:
  catalog.0004.alter_product_code
  Alter field code on product
  catalog.0005.rename_stock_product_quantity
  Rename field stock on product to quantity

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py sqlmigrate catalog 0005
BEGIN;
--
-- Rename field stock on product to quantity
--
ALTER TABLE "catalog_product" RENAME COLUMN "stock" TO "quantity";
COMMIT;

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, catalog, contenttypes, sessions
Running migrations:
  Applying catalog.0004.alter_product_code... OK
  Applying catalog.0005.rename_stock_product_quantity... OK

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
[X] 0001.initial
[X] 0002.add_product_code
[X] 0003.fill_product_codes
[X] 0004.alter_product_code
[X] 0005.rename_stock_product_quantity

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate catalog 0004
Operations to perform:
  Target specific migration: 0004.alter_product_code, from catalog
Running migrations:
  Rendering model states... DONE
  Unapplying catalog.0005.rename_stock_product_quantity... OK

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
[X] 0001.initial
[X] 0002.add_product_code
[X] 0003.fill_product_codes
[X] 0004.alter_product_code
[ ] 0005.rename_stock_product_quantity

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate catalog
Operations to perform:
  Apply all migrations: catalog
Running migrations:
  Applying catalog.0005.rename_stock_product_quantity... OK

(venv) C:\Users\ssosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>
```

Rollback to 0004 is shown, followed by the migration status.

11. Re-Apply 0005

```
python manage.py migrate catalog
```

Migration 0005 is applied again successfully. This completes the rollback and re-application test.

```

catalog.0004.alter_product_code
Alter field code on product

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py sqlmigrate catalog 0004
BEGIN;
-- Alter field code on product
CREATE TABLE "new_catalog_product" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "code" varchar(30) NOT NULL UNIQUE, "name" varchar(100) NOT NULL, "stock" integer NOT NULL);
INSERT INTO "new_catalog_product" ("id", "name", "stock", "code") SELECT "id", "name", "stock", coalesce("code", NULL) FROM "catalog_product";
DROP TABLE "catalog_product";
ALTER TABLE "new_catalog_product" RENAME TO "catalog_product";
COMMIT;

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py makemigrations catalog
Was product.stock renamed to product.quantity (a IntegerField)? [y/N] y
Migrations for 'catalog':
catalog/migrations/0005_rename_stock_product_quantity.py
- Rename field stock on product to quantity

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate --plan
Planned operations:
catalog.0004.alter_product_code
Alter field code on product
catalog.0005.rename_stock_product_quantity
Rename field stock on product to quantity

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py sqlmigrate catalog 0005
BEGIN;
--
-- Rename field stock on product to quantity
--
ALTER TABLE "catalog_product" RENAME COLUMN "stock" TO "quantity";
COMMIT;

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate
Operations to perform:
Apply all migrations: admin, auth, catalog, contenttypes, sessions
Running migrations:
  Applying catalog.0004.alter_product_code... OK
  Applying catalog.0005.rename_stock_product_quantity... OK

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
catalog
[X] 0001.initial
[X] 0002.add_product_code
[X] 0003.fill_product_codes
[X] 0004.alter_product_code
[X] 0005.rename_stock_product_quantity

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate catalog 0004
Operations to perform:
  Target specific migration: 0004.alter_product_code, from catalog
Running migrations:
  Rendering model states... DONE
  Unapplying catalog.0005.rename_stock_product_quantity... OK

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py showmigrations catalog
catalog
[X] 0001.initial
[X] 0002.add_product_code
[X] 0003.fill_product_codes
[X] 0004.alter_product_code
[ ] 0005.rename_stock_product_quantity

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>python manage.py migrate catalog
Operations to perform:
  Apply all migrations: catalog
Running migrations:
  Applying catalog.0005.rename_stock_product_quantity... OK

(env) C:\Users\sosyl\Desktop\WepBootcamp\python-bootcamp\Week 10\Day 1\migration_lab>

```

0005 is successfully re-applied.

12. Migration Timeline

- 0001_initial — Create Product with name and stock.
- 0002_add_product_code — Add code as nullable.
- 0003_fill_product_codes — Populate existing code values with RunPython.
- 0004_alter_product_code — Make code required and unique.
- 0005_rename_stock_product_quantity — Rename stock to quantity.

13. Key Commands

- makemigrations — create migration files from model changes.
- migrate — apply migrations to the database.
- showmigrations — check which migrations are applied.
- migrate --plan — preview migration operations.
- sqlmigrate — inspect the SQL generated for a migration.
- makemigrations --empty — create a custom/data migration.
- migrate catalog 0004 — move the app to a specific migration state.

14. Final Checklist

- ☒ Project and virtual environment created
- ☒ catalog registered in INSTALLED_APPS
- ☒ Product model created
- ☒ 0001_initial created
- ☒ 0002_add_product_code created
- ☒ Migration plan and SQL inspected
- ☒ 0003 data migration created
- ☒ Existing codes populated
- ☒ 0004 made code required and unique
- ☒ 0005 renamed stock to quantity
- ☒ All migrations applied
- ☒ Rollback to 0004 tested
- ☒ 0005 re-applied successfully